

Windowed fractional Fourier transform on graphs: Properties and fast algorithm

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ABSTRACT

A key challenge in the field of graph signal processing is to design proper transform methods to extract valuable information from signals on weighted graphs. This paper first defines convolution, translation and modulation operators in graph fractional domain, and proposes windowed graph fractional Fourier transform (WGFRFT). Related properties are proved. Then, a fast algorithm of WGFRFT is designed to increase flexibility. The robustness and superiority of the fast algorithm are verified via simulations with synthetic signals on graphs. Finally, an application to graph clustering is presented to show the practicability of WGFRFT.

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1. Introduction

Signals with topological features such as high dimensions, discreteness and irregularity are gaining prominence. Graphs become an important form of signal representation since they can be used to describe the geometry and topology of this kind of signal. In recent years, some fundamental signal processing methods have been extended to graph signals [1–6], giving rise to the graph-based transforms such as graph Fourier transform (GFT) [7,8], graph filters [9,10], sampling and recovery on graphs [11,12], fast algorithm [13–15]. These results have been applied to social networks, sensor networks [2], and extended to solve tasks appeared in machine learning and deep neural networks [4–6]. Graph-based transforms, in particular, are catching significant attention as they play a vital role in processing and analysis of graph signals. Graph signal processing (GSP) mainly includes two basic frameworks: (i) deriving from algebraic signal processing and using graph adjacency matrix as its building block [7]; (ii) deriving from spectral graph theory and using graph Laplacian matrix as its building block [16]. The latter framework studies the eigenvalues and eigenvectors of graph Laplacian matrix and its eigenbases are used to define the spectrum of the graph signal. Similar to the classical Fourier transform which could not capture the time-varying properties, GFT fails in vertex frequency representation. The windowed graph Fourier transform (WGFT) is proposed to solve this problem [17–22]. Different operators related to WGFT in graph domain are explored, such as convolution operator [17,19], modulation operator

[17,19], translation operator [17,23,24]. WGFT overcomes the drawback that weighted graphs with irregular structures lack a shift-invariant notion of translation. Thus it becomes a valuable tool to extract information from signals on weighted graphs and can illustrate the structural properties of data that are hidden in the vertex domain. Then the spectrogram generated during this process matches the intuition in classical discrete-time signal processing [17]. Another problem is that the classical Fourier transform cannot obtain local information. In order to obtain more detailed structural properties for graph signals, fractional order is also introduced to GSP [25–31]. Fractional graph signal processing is an emerging area. There are few research results about transform methods in graph fractional domain, which is a combination of fractional transform domain and graph spectral domain. Wang et al. proposed the definition of graph fractional Fourier transform (GFRFT) $\hat{f} = P^\alpha f$ based on graph adjacency matrix $A = V \Lambda_A V^{-1}$ where $P = V^{-1}$ [29,30]. The definition of the GFRFT matrix parallels to the definition of discrete fractional Fourier transform matrix to the α order. More related contents can be found in [29]. Wu et al. proposed a new spectral graph fractional Fourier transform (SGFRFT) [31]. GFRFT and SGFRFT have shown advantages for revealing local characteristics of graph signals.

To analyze graph signal in vertex frequency domain, we propose a new transform. As an extended version of our conference paper [32], we define the graph fractional convolution, translation, and modulation operators and present their detailed properties. Then we define windowed graph fractional Fourier transform (WGFRFT) based on these new operators. An inverse transform is also given. Furthermore, we propose a fast algorithm and carry out simulations with synthetic signals on graphs. The performance between the original method and our scheme is compared by ver-

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tex frequency representations, root mean square errors, and time costs. The results show that our algorithm brings an outstanding improvement of computational complexity and robustness. It further demonstrates that our method is more suitable for processing contaminated signals than the original method. Finally, we apply WGRFT to graph clustering to clarify its application potentials.

2. Preliminaries

2.1. Spectral graph theory

Recall that in the continuous setting, let $\mathcal{F} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ be the Fourier transform. Then we have

$$\mathcal{F}(\Psi_n) = e^{-in\pi/2} \Psi_n,$$

where Ψ_n is the n th Hermite-Gauss function. The fractional Fourier transform shares the same eigenfunctions as Fourier transform but raises the eigenvalues to the fractional power. That is, for any $\alpha \in \mathbb{R}$, we have

$$\mathcal{F}^\alpha \Psi_n = e^{-i\alpha n\pi/2} \Psi_n.$$

This definition of fractional Fourier transform motivates researchers to define its discrete version on the graph setting in the following way: first we define GFT and find its eigenvalues and eigenvectors [16]. Then we retain the eigenvectors and raise the eigenvalues to the fractional power [31].

An undirected weighted graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, W\}$ consists of a finite set of vertices $\mathcal{V} = \{v_0, \dots, v_{N-1}\}$, where $N = |\mathcal{V}|$ is the number of nodes, a set of edges $\mathcal{E}$, and a weighted adjacency matrix W . The non-normalized graph Laplacian is a symmetric difference operator $\mathcal{L} = D - W$ [16], where D is a diagonal degree matrix of $\mathcal{G}$. Let $\{\chi_0, \chi_1, \dots, \chi_{N-1}\}$ be the set of orthonormal eigenvectors. Suppose that the corresponding Laplacian eigenvalues are sorted as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1} := \lambda_{\max}$. Therefore

$$\mathcal{L} = \chi \Lambda \chi^H, \quad (1)$$

where χ is the unitary column matrix

$$\chi = [\chi_0, \chi_1, \dots, \chi_{N-1}], \quad (2)$$

and the diagonal matrix is $\Lambda = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{N-1}])$, the superscript H represents the Hermitian transpose operation. For connected graphs, the eigenvector χ_0 associated with the eigenvalue 0 is constant and equals $(\frac{1}{\sqrt{N}})$ at each vertex.

A signal defined on the graph $\mathcal{G}$ is a map [29]

$$f : \mathcal{V} \rightarrow \mathbb{R}, \\ v_n \mapsto f(n),$$

and f can also be written as a real-valued vector

$$f = [f(0), f(1), \dots, f(N-1)]^T \in \mathbb{R}^N.$$

We adopt the definition of (inverse) GFT as in [16]. The GFT of f is

$$\hat{f}(\ell) = \langle f, \chi_\ell \rangle = \sum_{n=0}^{N-1} f(n) \chi_\ell^*(n), \ell = 0, 1, \dots, N-1. \quad (3)$$

The inverse GFT is given by

$$f(n) = \sum_{\ell=0}^{N-1} \hat{f}(\ell) \chi_\ell(n), n = 0, 1, \dots, N-1. \quad (4)$$

Thus we see that the GFT is given by the matrix χ . According to our discussion in the beginning, we need to apply α th power to its eigenvalues and retain its eigenvectors. The process is equivalent to the power function of matrices. Thus SGFRFT is given by χ^α . The graph fractional Laplacian operator $\mathcal{L}_\alpha$ is defined by $\mathcal{L}_\alpha = \kappa R \kappa^H$, where $0 < \alpha \leq 1$, $\ell = 0, 1, \dots, N-1$ [31]. Note that as χ is unitary then

$$\kappa = [\kappa_0, \kappa_1, \dots, \kappa_{N-1}] = \chi^\alpha, \quad (5)$$

is also unitary. And

$$R = \text{diag}([r_0, r_1, \dots, r_{N-1}]) = \Lambda^\alpha, \quad (6)$$

so that

$$r_\ell = \lambda_\ell^\alpha. \quad (7)$$

The SGFRFT of any signal f building on the graph $\mathcal{G}$ is defined by [31]:

$$\hat{f}_\alpha(\ell) = \langle f, \kappa_\ell \rangle = \sum_{n=0}^{N-1} f(n) \kappa_\ell^*(n), \ell = 0, 1, \dots, N-1, \quad (8)$$

when $\alpha = 1$, the SGFRFT degenerates into standard GFT.

The inverse SGFRFT is given by

$$f(n) = \sum_{\ell=0}^{N-1} \hat{f}_\alpha(\ell) \kappa_\ell(n), n = 0, 1, \dots, N-1. \quad (9)$$

The Parseval relation of the SGFRFT holds, for any signal f and g defined on the graph $\mathcal{G}$ we have:

$$\langle f, g \rangle = \langle \hat{f}_\alpha, \hat{g}_\alpha \rangle. \quad (10)$$

If $f = g$, then

$$\begin{aligned} \sum_{n=0}^{N-1} |f(n)|^2 &= \|f\|_2^2 = \langle f, f \rangle \\ &= \langle \hat{f}_\alpha, \hat{f}_\alpha \rangle = \|\hat{f}_\alpha\|_2^2 = \sum_{\ell=0}^{N-1} |\hat{f}_\alpha(\ell)|^2. \end{aligned} \quad (11)$$

2.2. Windowed fractional Fourier transform

Given a signal $f \in L^2(\mathbb{R})$, a window function $\psi \in L^2(\mathbb{R})$, the windowed fractional Fourier transform (WFRFT) of f is defined as [26]

$$(S_\psi f)(u, \xi) = \int_{\mathbb{R}} f(t) \psi^*(t-u) K_\alpha(t, \xi) dt, \quad (12)$$

where

$$K_\alpha(t, \xi) = \begin{cases} A_\alpha \exp\{j\frac{\xi^2}{2} \cot(\alpha) - jt\xi \csc(\alpha) + j\frac{t^2}{2} \cot(\alpha)\}, & \alpha \neq n\pi \\ \delta(\xi - t), & \alpha = 2n\pi \\ \delta(\xi + t), & \alpha = 2n\pi - \pi \end{cases} \quad (13)$$

is the kernel of the fractional Fourier transform (FRFT). The constants are

$$A_\alpha = \sqrt{\frac{1 - j \cot(\alpha)}{2\pi}}, \alpha = \frac{p\pi}{2}, 0 < |p| < 2, 0 < |\alpha| < \pi.$$

Then, we give a more intuitive view of the WFRFT. For any function $f \in L^2(\mathbb{R})$ and $u \in \mathbb{R}$, the translation operator $T_u : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ is defined by

$$(T_u f)(t) = f(t - u), \quad (14)$$

let $h(t, u) = (f T_u \psi^*)(t)$, so we have

$$(S_\psi f)(u, \xi) = \int_{\mathbb{R}} h(t, u) K_\alpha(t, \xi) dt. \quad (15)$$

Therefore, the WFRFT of f is equal to the FRFT of h .

In practical applications involving large graphs, we are more concerned about the local features of graphs. Therefore, it is necessary to characterize local signal features in the joint vertex-frequency domain. Introducing window functions or fractional orders in transforms can bring local analysis to signals. So we design a transform that taking the fractional Laplacian matrix and window function into account. To obtain more detailed structural properties for graph signal, the windowed graph fractional Fourier transform (WGFRFT) and related properties are proposed.

3. Windowed graph fractional Fourier transform

3.1. Operators in graph fractional domain

Similar to WFRFT, defining the window transform in graph fractional domain needs new translation and modulation operators. Classical translation operator in (14) can be interpreted as the convolution of signal f and Delta function. Therefore, to define the translation operator we first define the convolution operator in graph fractional domain.

Convolution in the time domain is equivalent to multiplication in the Fourier domain. In fractional transform domain, there are several kinds of convolution operators [33,34]. In graph fractional domain, by exploiting the graph fractional Laplacian eigenvectors, the graph fractional convolution operator is defined as the following form which is consistent with the classical convolution.

Definition 1 (Graph fractional convolution operator). For any graph signal f, g , their graph fractional convolution $*_\alpha$ is

$$(f *_\alpha g)(n) = \sum_{\ell=0}^{N-1} \widehat{f}_\alpha(\ell) \widehat{g}_\alpha(\ell) \kappa_\ell(n). \quad (16)$$

Based on this definition, it is easy to prove that the graph fractional convolution operator satisfies the following properties.

Lemma 1. (i) Convolution in the vertex domain is multiplication in graph fractional domain.

$$(\widehat{f *_\alpha g})_\alpha = \widehat{f}_\alpha \widehat{g}_\alpha. \quad (17)$$

(ii) $\forall a \in \mathbb{R}$,

$$a(f *_\alpha g) = (af) *_\alpha g = f *_\alpha (ag). \quad (18)$$

(iii) Commutativity:

$$f *_\alpha g = g *_\alpha f. \quad (19)$$

(iv) Distributivity: given a function $h \in \mathbb{R}^N$ be arbitrary,

$$f *_\alpha (g + h) = f *_\alpha g + f *_\alpha h. \quad (20)$$

(v) Associativity:

$$(f *_\alpha g) *_\alpha h = f *_\alpha (g *_\alpha h). \quad (21)$$

(vi) Identity element: given a function $g_0 \in L^2(\mathbb{R})$ and $g_0(n) := \sum_{\ell=0}^{N-1} \kappa_\ell(n)$, then g_0 is an identity for convolution operator $*_\alpha$,

$$f *_\alpha g_0 = f. \quad (22)$$

To shift a window in vertex domain, we give a graph fractional translation operator $T_i^\alpha : \mathbb{R}^N \rightarrow \mathbb{R}^N$ via graph fractional convolution with a Delta function $\delta_i(n)$ centered at vertex v_i .

Definition 2 (Graph fractional translation operator). For any signal f defined on the graph $\mathcal{G}$ and any $i \in \{0, 1, \dots, N-1\}$, the graph fractional translation T_i^α is defined as

$$(T_i^\alpha f)(n) = (\sqrt{N})^\alpha (f *_\alpha \delta_i)(n), \quad (23)$$

where

$$\delta_i(n) = \begin{cases} 1, & \text{if } n = i \\ 0, & \text{otherwise.} \end{cases} \quad (24)$$

Computing the $*_\alpha$ in (23), we obtain

$$(T_i^\alpha f)(n) = (\sqrt{N})^\alpha \sum_{\ell=0}^{N-1} \widehat{f}_\alpha(\ell) \kappa_\ell^*(i) \kappa_\ell(n). \quad (25)$$

From properties of graph fractional convolution operator, we can directly derive basic properties of graph fractional translation operator.

Lemma 2. For any $f, g \in \mathbb{R}^N$ and $i, j \in \{1, 2, \dots, N\}$

- (i) $T_i^\alpha (f *_\alpha g) = (T_i^\alpha f) *_\alpha g = f *_\alpha (T_i^\alpha g)$,
- (ii) $T_i^\alpha T_j^\alpha f = T_j^\alpha T_i^\alpha f$.

Proof.

- (i) By using the commutativity and associativity of graph fractional convolution operator, we have

$$\begin{aligned} T_i^\alpha (f *_\alpha g) &= (\sqrt{N})^\alpha ((f *_\alpha g) *_\alpha \delta_i) \\ &= (\sqrt{N})^\alpha (f *_\alpha (g *_\alpha \delta_i)) \\ &= (f *_\alpha ((\sqrt{N})^\alpha g *_\alpha \delta_i)) \\ &= f *_\alpha (T_i^\alpha g). \end{aligned}$$

Similarly we can prove that $T_i^\alpha (f *_\alpha g) = (T_i^\alpha f) *_\alpha g$;

- (ii) From the definition of graph fractional translation operator,

$$\begin{aligned} T_i^\alpha T_j^\alpha f &= (\sqrt{N})^{2\alpha} ((f *_\alpha \delta_j) *_\alpha \delta_i) \\ &= (\sqrt{N})^{2\alpha} ((f *_\alpha \delta_i) *_\alpha \delta_j) \\ &= T_j^\alpha T_i^\alpha f. \end{aligned}$$

Thus the graph fractional translation operator is commutative.

According to the above definition, we find an useful property about the bounds of the graph fractional translation operator.

For the simplicity of writing later, first define the largest absolute value of the elements of a given graph fractional Laplacian eigenvector as

$$a_\ell = \|\kappa_\ell\|_\infty = \max_{i \in \{1, 2, \dots, N\}} |\kappa_\ell(i)|, \quad (26)$$

and

$$b_i = \max_{\ell \in \{0,1,\dots,N-1\}} |\kappa_\ell(i)|. \quad (27)$$

Note that

$$M = \max_{\ell \in \{0,1,\dots,N-1\}} \{a_\ell\} = \max_{i \in \{1,2,\dots,N\}} \{b_i\}. \quad (28)$$

Lemma 3. For any $f \in \mathbb{R}^N$,

$$\|T_i^\alpha f\|_2 \leq (\sqrt{N})^\alpha b_i \|\widehat{f}_\alpha\|_2 \leq (\sqrt{N})^\alpha M \|\widehat{f}_\alpha\|_2. \quad (29)$$

Proof.

$$\begin{aligned} \|T_i^\alpha f\|_2^2 &= \sum_{n=1}^N \left((\sqrt{N})^\alpha \sum_{\ell=0}^{N-1} \widehat{f}_\alpha(\ell) \kappa_\ell^*(i) \kappa_\ell(n) \right)^2 \\ &= N^\alpha \sum_{\ell=0}^{N-1} \sum_{\ell'=0}^{N-1} \widehat{f}_\alpha(\ell) \widehat{f}_\alpha(\ell') \kappa_\ell^*(i) \kappa_{\ell'}^*(i) \sum_{n=1}^N \kappa_\ell(n) \kappa_{\ell'}(n) \end{aligned} \quad (30)$$

$$= N^\alpha \sum_{\ell=0}^{N-1} |\widehat{f}_\alpha(\ell)|^2 |\kappa_\ell^*(i)|^2 \quad (31)$$

$$\leq N^\alpha b_i^2 \|\widehat{f}_\alpha\|_2^2. \quad (32)$$

By (28), the last inequality in (29) holds.

Definition 3 (Graph fractional modulation operator). For any signal f defined on the graph $\mathcal{G}$ and any $k \in \{0, 1, \dots, N-1\}$, define the graph fractional modulation M_k^α by

$$(M_k^\alpha f)(n) = (\sqrt{N})^\alpha f(n) \kappa_k(n). \quad (33)$$

3.2. Windowed graph fractional Fourier transform

Based on the preliminaries and the above operators in graph fractional domain, a novel windowed graph fractional Fourier transform is defined by the following Definition 4.

Definition 4 (Windowed graph fractional Fourier transform). Given a graph signal f , a window function $\psi \in L^2(\mathbb{R})$, the WGRFT $(W_{\psi,\alpha} f)(i, \ell)$ of f is

$$(W_{\psi,\alpha} f)(i, \ell) = N^\alpha \sum_{d=1}^N f(d) \left(\sum_{m=0}^{N-1} \widehat{\psi}_\alpha^*(m) \kappa_m(i) \kappa_m^*(d) \right) \kappa_i^*(\ell). \quad (34)$$

Remark 1. Note that $\kappa = \chi^\alpha$, and let

$$\begin{aligned} \Psi_{i,\ell}(n) &= (M_\ell^\alpha T_i^\alpha \psi)(n) \\ &= N^\alpha \kappa_\ell(n) \sum_{m=0}^{N-1} \widehat{\psi}_\alpha(m) \kappa_m^*(i) \kappa_m(n), \end{aligned} \quad (35)$$

then

$$(W_{\psi,\alpha} f)(i, \ell) = \langle f, \Psi_{i,\ell} \rangle. \quad (36)$$

Also note that the proposed WGRFT in (36) degenerates to the WGFT defined by Shuman in [17] when $\alpha = 1$.

Remark 2. We can give an intuitive interpretation to WGRFT. Let $h_i(n) = (\sqrt{N})^\alpha (f(T_i^\alpha \psi)^*)(n)$, then

$$(W_{\psi,\alpha} f)(i, \ell) = \sum_{n=1}^N h_i(n) \kappa_\ell^*(n) = \widehat{h_{i,\alpha}}(\ell). \quad (37)$$

Therefore $(W_{\psi,\alpha} f)(i, \ell)$ is equal to the SGFRFT of $h_i(n)$.

We show a simulation example of the WGRFT by modifying the GSPBOX toolbox [35] to present a visualization and qualitative analysis of our method. We perform the proposed WGRFT on a synthetic Swiss roll data. The Swiss roll graph is generated by randomly sampling 500 points from a two-dimensional manifold. Then we assign edge weights to this graph based on the Euclidean distances and Gaussian kernel

$$W_{ij} = \begin{cases} \exp(-\frac{[d(i,j)]^2}{2\sigma_1^2}) & \text{if } d(i,j) \leq \sigma_2 \\ 0 & \text{otherwise,} \end{cases}$$

and the signal f on graph is

$$f = \sin(2\pi n/N), \quad 1 \leq n \leq 500.$$

We use $\sigma_1 = 0.100, \sigma_2 = 0.215$ for computing the Swiss roll graph, and select Gaussian window as the window function. Fig. 1 shows the original data set with Fig. 1(a), and some WGRFT coefficients for different fractional order preforming on this underlying weighted graph with Fig. 1(b)-Fig. 1(d). Among them, Fig. 1(b) represents the parameter $\alpha = 1$, which is also the result of WGFT. Compared to WGFT, WGRFT introduces another new parameter to balance or enhance the transform coefficients among different vertices, thus making the local analysis more flexible. By tuning the fractional order, WGRFT provides more information hidden in graph fractional domain that classical methods could not explore.

4. Transform properties

In this section, we will show that the windowed graph fractional Fourier transform admits an inverse formula. We provide a property of the window function. The existence of this property guarantees the reasonableness of the inverse transform.

4.1. A property of the window function

Theorem 1. If a window function $\psi \in L^2(\mathbb{R})$ satisfies $T_i^\alpha \psi \neq 0$, then the family $\{\Psi_{i,\ell}\}_{i=1,2,\dots,N; \ell=0,1,\dots,N-1}$ forms a frame, thus for any signal $f \in \mathbb{R}^N$

$$0 < A \|f\|_2^2 \leq \sum_{i=1}^N \sum_{\ell=0}^{N-1} |\langle f, \Psi_{i,\ell} \rangle|^2 \leq B \|f\|_2^2 < \infty, \quad (38)$$

where

$$0 < A = \min_{n \in \{1,2,\dots,N\}} \{N^\alpha \|T_n^\alpha \psi\|_2^2\},$$

and

$$B = \max_{n \in \{1,2,\dots,N\}} \{N^\alpha \|T_n^\alpha \psi\|_2^2\} < \infty.$$

Proof. Combining (36) and (34),

$$\begin{aligned} \sum_{i=1}^N \sum_{\ell=0}^{N-1} |\langle f, \Psi_{i,\ell} \rangle|^2 &= \sum_{i=1}^N \sum_{\ell=0}^{N-1} |(W_{\psi,\alpha} f)(i, \ell)|^2 \\ &= \sum_{i=1}^N \sum_{\ell=0}^{N-1} |\widehat{h_{i,\alpha}}(\ell)|^2. \end{aligned} \quad (39)$$

By (11),

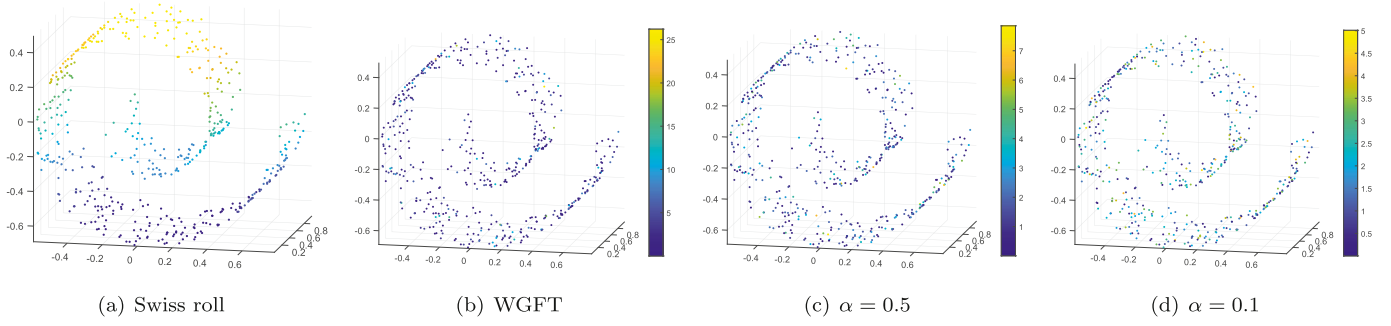


Fig. 1. WGRFT on Swiss roll data with different fractional order α . The first figure presents the original Swiss roll data cloud. The second figure shows WGFT of this data, which is equivalent to fractional order $\alpha = 1$. The last two pictures present WGRFT with $\alpha = 0.5$ and $\alpha = 0.1$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$\begin{aligned}
 \sum_{i=1}^N \sum_{\ell=0}^{N-1} |\langle f, \Psi_{i,\ell} \rangle|^2 &= \sum_{i=1}^N \sum_{n=1}^N |h_i(n)|^2 \\
 &= N^\alpha \sum_{i=1}^N \sum_{n=1}^N |f(n)|^2 |T_i^\alpha \psi(n)|^2 \\
 &= N^\alpha \sum_{i=1}^N \sum_{n=1}^N |f(n)|^2 |T_n^\alpha \psi(i)|^2 \\
 &= N^\alpha \sum_{n=1}^N |f(n)|^2 \|T_n^\alpha \psi\|_2^2.
 \end{aligned} \tag{40}$$

Let

$$A = \min_{n \in \{1, 2, \dots, N\}} \{N^\alpha \|T_n^\alpha \psi\|_2^2\},$$

and

$$B = \max_{n \in \{1, 2, \dots, N\}} \{N^\alpha \|T_n^\alpha \psi\|_2^2\},$$

using Lemma 3, we have

$$A > 0, \tag{41}$$

and

$$B \leq N^{2\alpha} M^2 \|\psi\|_2^2 < \infty. \tag{42}$$

Hence

$$0 < A \|f\|_2^2 \leq \sum_{i=1}^N \sum_{\ell=0}^{N-1} |\langle f, \Psi_{i,\ell} \rangle|^2 \leq B \|f\|_2^2 < \infty. \tag{43}$$

4.2. Reconstruction formula

If the window function ψ satisfies the condition in Theorem 1, a reconstruction formula holds.

Theorem 2. If ψ satisfies $T_i^\alpha \psi \neq 0$, then for any signal $f \in \mathbb{R}^N$,

$$f(n) = \frac{1}{N^\alpha \|T_i^\alpha \psi\|_2^2} \sum_{i=1}^N \sum_{\ell=0}^{N-1} (W_{\psi,\alpha} f)(i, \ell) \Psi_{i,\ell}(n). \tag{44}$$

Proof. By Definition 4 we have

$$\sum_{i=1}^N \sum_{\ell=0}^{N-1} (W_{\psi,\alpha} f)(i, \ell) \Psi_{i,\ell}(n)$$

$$\begin{aligned}
 &= \sum_{i=1}^N \sum_{\ell=0}^{N-1} \left(N^\alpha \sum_{d=1}^N f(d) \left(\sum_{m=0}^{N-1} \hat{\psi}_\alpha^*(m) \kappa_m(i) \kappa_m^*(d) \right) \kappa_i^*(d) \right) \\
 &\quad \times \left(N^\alpha \kappa_i(n) \sum_{j=0}^{N-1} \hat{\psi}_\alpha(j) \kappa_j^*(i) \kappa_j(n) \right) \\
 &= N^{2\alpha} \sum_{d=1}^N f(d) \times \sum_{m=0}^{N-1} \sum_{j=0}^{N-1} \hat{\psi}_\alpha^*(m) \hat{\psi}_\alpha(j) \kappa_m^*(d) \kappa_j(n) \\
 &\quad \times \sum_{i=1}^N \kappa_m(i) \kappa_j^*(i) \sum_{l=0}^{N-1} \kappa_l^*(d) \kappa_l(n) \\
 &= N^{2\alpha} \sum_{d=1}^N f(d) \sum_{m=0}^{N-1} \sum_{j=0}^{N-1} \hat{\psi}_\alpha^*(m) \hat{\psi}_\alpha(j) \kappa_m^*(d) \kappa_j(n) \delta_{mj} \delta_{dn} \\
 &= N^{2\alpha} f(n) \sum_{m=0}^{N-1} |\hat{\psi}_\alpha(m)|^2 |\kappa_m(n)|^2 \\
 &= N^\alpha f(n) \|T_i^\alpha \psi\|_2^2.
 \end{aligned}$$

By Lemma 3, the last equality obtained.

From the definition (34) of WGRFT, it is not difficult to see that, like WGFT, WGRFT also needs high computational complexity. First for a graph with N vertices, to compute $(W_{\psi,\alpha} f)(i, \ell)$ defined as (34), it requires $O(N^2)$ operations. Then we have N^2 elements for $W_{\psi,\alpha} f$. Thus the global complexity is $O(N^4)$. Such a high computational complexity is not desirable in applications. Thus, we propose the following fast algorithm for WGRFT, which can effectively shorten the computational time.

5. A fast algorithm for WGRFT

5.1. Fast windowed graph fractional Fourier transform

In order to improve the calculation speed, a fast algorithm of WGRFT is proposed in this section. In the classical case of the one-dimensional signal, the windowed Fourier transform can be calculated by performing the inverse Fourier transform from the ' α -domain' [36]. The α -domain is defined as the Fourier transform of the windowed Fourier domain.

Analogously to the above calculation, we define ' Θ -domain' in fractional Fourier domain by

$$\Theta(\xi, \xi') = \int_{-\infty}^{+\infty} (S_\psi f)(u, \xi) K_\alpha(u, \xi') du, \tag{45}$$

where $(S_\psi f)(u, \xi)$ is WFRFT of f in (12), that is, (45) applies FRFT to $S_\psi f$. The fast windowed fractional Fourier transform (FWFRFT) can be achieved by applying inverse FRFT to Θ on the variable ξ' .

$$(FS_\psi f)(u, \xi) = \int_{-\infty}^{+\infty} \Theta(\xi, \xi') K_{-\alpha}(u, \xi') d\xi'. \quad (46)$$

Recall that $S_\psi f$ is

$$(S_\psi f)(u, \xi) = \langle f, M_\xi T_u \psi \rangle, \quad (47)$$

where $(M_\xi g)(t) = g(t) K_{-\alpha}(t, \xi)$.

Applying the above idea to graphs, we define Θ -domain as the SGFRFT of WGRFT in graph fractional domain:

$$\Theta(\ell, \ell') = \sum_{n=1}^N (W_{\psi, \alpha} f)(n, \ell') \kappa_\ell^*(n). \quad (48)$$

Then, the discretization of Θ is

$$\begin{aligned} & \Theta(\ell, \ell') \\ &= \sum_{n=1}^N (W_{\psi, \alpha} f)(n, \ell') \kappa_\ell^*(n) \\ &= \sum_{n=1}^N \left(N^\alpha \sum_{d=1}^N f(d) \left(\sum_{m=0}^{N-1} \hat{\psi}_\alpha^*(m) \kappa_m(n) \kappa_m^*(d) \right) \kappa_{\ell'}^*(d) \right) \kappa_\ell^*(n) \\ &= N^\alpha \sum_{m=0}^{N-1} \sum_{d=1}^N f(d) \kappa_{\ell'}^*(d) \kappa_m^*(d) \hat{\psi}_\alpha^*(m) \left(\sum_{n=1}^N \kappa_m(n) \kappa_\ell^*(n) \right) \\ &= N^\alpha \sum_{m=0}^{N-1} \sum_{d=1}^N f(d) \kappa_{\ell'}^*(d) \kappa_m^*(d) \hat{\psi}_\alpha^*(m) \delta_{m\ell} \\ &= N^\alpha \sum_{d=1}^N f(d) \kappa_{\ell'}^*(d) \kappa_\ell^*(d) \hat{\psi}_\alpha^*(\ell). \end{aligned} \quad (49)$$

Define $\tilde{f}(\ell, \ell') = \sum_{d=1}^N f(d) \kappa_{\ell'}^*(d) \kappa_\ell^*(d)$. We get

$$\Theta(\ell, \ell') = N^\alpha \tilde{f}(\ell, \ell') \hat{\psi}_\alpha^*(\ell). \quad (50)$$

Definition 5 (Fast windowed graph fractional Fourier transform). The inverse SGFRFT of Θ on the variable ℓ is the fast windowed graph fractional Fourier transform (FWGRFT).

$$\begin{aligned} (FW_{\psi, \alpha} f)(n, \ell') &= \sum_{\ell=0}^{N-1} \Theta(\ell, \ell') \kappa_\ell(n) \\ &= \sum_{\ell=0}^{N-1} N^\alpha \tilde{f}(\ell, \ell') \hat{\psi}_\alpha^*(\ell) \kappa_\ell(n). \end{aligned} \quad (51)$$

5.2. Algorithm for calculating FWGRFT

The algorithm for calculating FWGRFT is defined through the following steps:

- (i) Calculate an $N \times N$ matrix which represents:

$$\tilde{f}(\ell, \ell') = \sum_{d=1}^N f(d) \kappa_{\ell'}^*(d) \kappa_\ell^*(d). \quad (52)$$

This step has $O(N^3)$ computational complexity. One can calculate $\tilde{f}(\ell, \ell')$ by using matrices:

$$\tilde{F} = [\tilde{f}(\ell, \ell')] = (F \circ \kappa^H) \cdot \kappa, \quad (53)$$

where $\circ$ denotes Hadamard multiplication, i.e. element-wise multiplication of two matrices, and $\cdot$ represents the standard matrix multiplication. The matrix F is an $N \times N$ matrix such that each row of F is the signal f , and κ^H denotes the complex conjugate of the graph fractional basis κ .

- (ii) Form a matrix such that each row is the SGFRFT of the window function.

$$\Psi^\alpha = \begin{pmatrix} \hat{\psi}_\alpha(r_0) & \hat{\psi}_\alpha(r_1) & \cdots & \hat{\psi}_\alpha(r_{N-1}) \\ \hat{\psi}_\alpha(r_0) & \hat{\psi}_\alpha(r_1) & \cdots & \hat{\psi}_\alpha(r_{N-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\psi}_\alpha(r_0) & \hat{\psi}_\alpha(r_1) & \cdots & \hat{\psi}_\alpha(r_{N-1}) \end{pmatrix} \quad (54)$$

This step does not influence the computational complexity.

- (iii) Calculate the Θ domain representation by multiplying $\tilde{F}$ and $(\Psi^\alpha)^*$.

$$\Theta = N^\alpha (\Psi^\alpha)^* \cdot \tilde{F}. \quad (55)$$

The computational complexity is the complexity for matrix multiplication, i.e. $O(N^3)$.

- (iv) Finally, the WGRFT at each vertex is obtained by applying inverse SGFRFT to Θ , i.e.

$$W_{\psi, \alpha} f = \kappa \cdot \Theta. \quad (56)$$

Thus, the complexity of this step is $O(N^3)$.

To sum up, the computational complexity for FWGRFT is $O(N^3)$. As N becomes larger, this computational complexity is a significant improvement compared with the original algorithm which has $O(N^4)$ computational complexity. In the next part, we apply the algorithm and show time cost to reflect the superiority of our scheme.

Remark 3. The term “fast algorithm” here specifically refers to the reduction of $O(N^4)$ to $O(N^3)$, and should not be confused with other fast algorithms in the signal processing literature.

5.3. Algorithm simulation

We carry out two basic simulations to show the efficiency of WGRFT and FWGRFT. In the first experiment we show the vertex frequency representations of three different time series based on path graph, which were presented in Fig. 2. In the other experiment we show the vertex frequency representations by using the graph fractional Laplacian eigenvectors as the graph signal based on three different graph structures, which were presented in Fig. 4. From Fig. 2 and Fig. 4 we see that the frequencies obtained by the fast algorithm are consistent with the original transform and are more obvious than their original performance, which verifies the correctness of the algorithm.

- (i) Simulation 1

First, given a path graph with trivial edge weights, we consider the following three time series: signal f_1 is a sampled sinusoidal signal with only one frequency, f_2 is a sampled sinusoidal signal with three different frequencies, f_3 is a chirp signal (see them in (57), (58), (59)).

$$f_1(n) = \sin(80\pi n/N), \quad 1 \leq n \leq 180 \quad (57)$$

$$f_2(n) = \begin{cases} \sin(90\pi n/N), & 1 \leq n \leq 60 \\ \sin(60\pi n/N), & 60 < n \leq 120 \\ \sin(120\pi n/N), & 120 < n \leq 180 \end{cases} \quad (58)$$

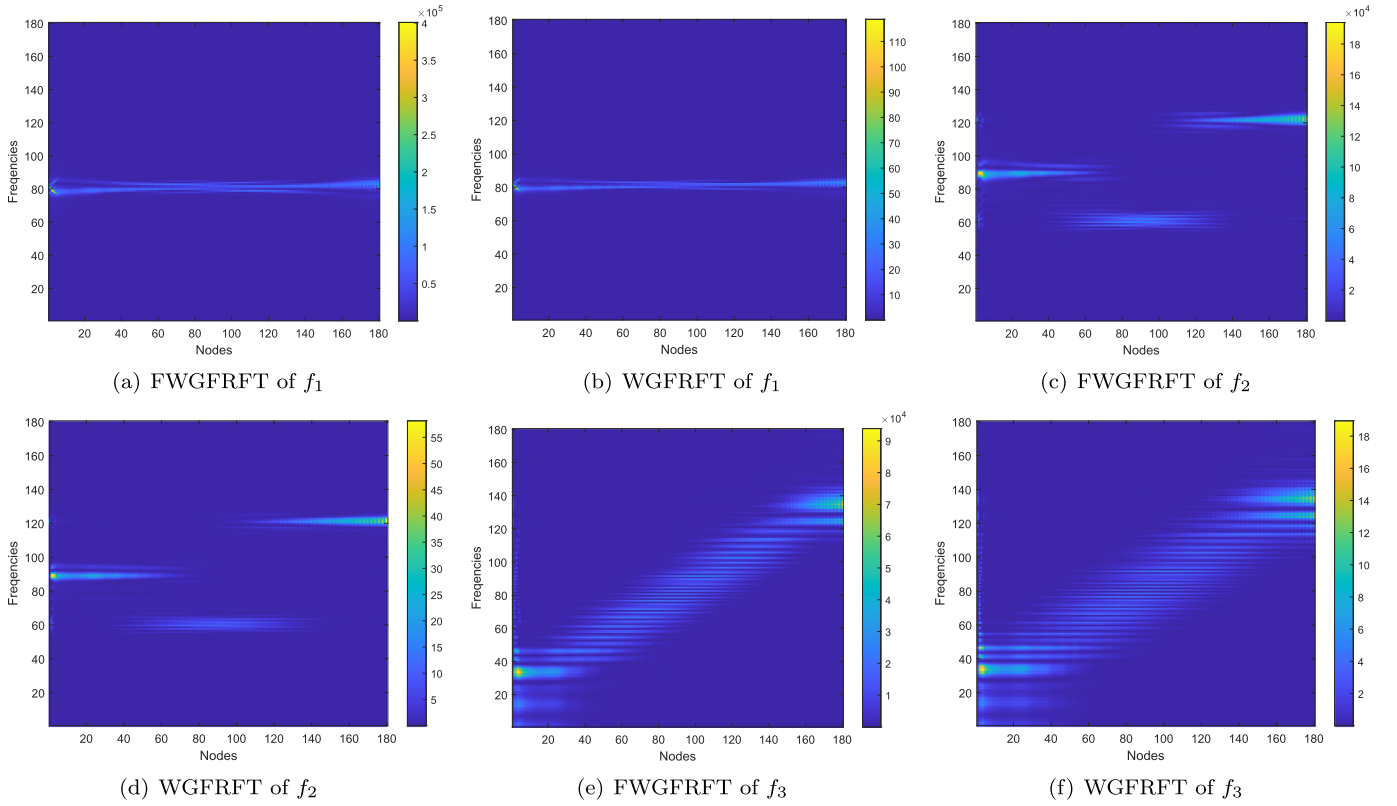


Fig. 2. Vertex frequency representations of signals f_1 , f_2 , and f_3 by FWGRFT and WGRFT, with fractional order $\alpha = 0.9$.

$$f_3(n) = \sin\left(\left(20n + \frac{1}{3}n^2\right)\pi/N\right), 1 \leq n \leq 180 \quad (59)$$

Fig. 2 displays the vertex frequency representation of signals f_1 , f_2 and f_3 . We select the standard Gauss function as window function ψ in (34) and (51). Next we apply SGFRFT to the window function with fractional order $\alpha = 0.9$ and then normalize, i.e. we obtain $\hat{\psi}_{0.9}$. For WGRFT, we apply the defining equation (34) to our algorithm, whose complexity is $O(N^4)$; for FWGRFT, we use the algorithm discussed in section 5.

What we can conclude from Fig. 2 is that, for the first signal f_1 , we see from Fig. 2(a) and Fig. 2(b) that our algorithms effectively find out the correct frequency. Similarly, for the second signal f_2 , Fig. 2(c) and Fig. 2(d) show that three different frequencies are successfully extracted and separated. For the chirp signal f_3 , Fig. 2(e) and Fig. 2(f) correctly present the increasing frequency of up-chirp.

Finally, we remark that unlike GFT on special graphs such as line graphs and circular graphs, GFRFT does not descend to any traditional discrete transforms. A possible explanation is that we cannot fix an eigendecomposition of the GFT matrix. Therefore, the GFRFT matrix is not as regular as GFT. The form becomes more complex with the participation of the discrete window function. However, the simulation results are sound to show the effectiveness of WGRFT.

(ii) Simulation 2

In the second experiment, we simulate three different graphs: a 300 nodes Swiss roll graph, a 256 nodes Community graph, and a 128 nodes Sensor network. We consider the real part of three graph signal as:

$$f_4(n) = \text{Re}(\kappa_{60}^{\text{sr}}(n))$$

$$f_5(n) = \text{Re}(\kappa_{64}^{\text{c}}(n) + \kappa_{128}^{\text{c}}(n))$$

$$f_6(n) = \text{Re}(\kappa_{10}^{\text{s}}(n) + \kappa_{50}^{\text{s}}(n) + \kappa_{90}^{\text{s}}(n))$$

As a signal on the Swiss roll graph, we use $\kappa_{60}(n)$ from the graph fractional Laplacian matrix (5) of the Swiss roll graph. As a signal on the Community graph, we use the summation of $\kappa_{64}(n)$ and $\kappa_{128}(n)$ from the graph fractional Laplacian of the Community graph. As a signal on the Sensor network graph, we use the summation of $\kappa_{10}(n)$, $\kappa_{50}(n)$, and $\kappa_{90}(n)$ from the graph fractional Laplacian of the Sensor network graph. The fractional order in these graph fractional Laplacian eigenvectors is $\alpha = 0.9$. Details of the original signal are given in Fig. 3.

Fig. 4 displays the vertex frequency representation of signals f_4 , f_5 and f_6 . We choose the same window function as in the first experiment. From Fig. 4, we can see that our algorithms effectively find out the correct frequencies. The frequency representations of FWGRFT are clearer than those of WGRFT, which further verify the correctness and effectiveness of our fast algorithm.

5.4. Algorithm evaluation

In order to evaluate the performance of the fast algorithm, we carry out experiments to show both robustness and efficiency. To test robustness, we apply the fast algorithm to signals polluted by noises. To test efficiency, we compare time costs of FWGRFT and WGRFT on the same data set.

(i) Robustness

We verify the robustness of our algorithm from both quantitative and qualitative aspects. In the first experiment, we fix the fractional order and change the noise variances on different signals. After FWGRFT, we visualize the results by presenting the vertex frequency representations. In the second experiment, we change both fractional orders and noise variances. We compute the root mean square error between transformation coefficients of original signals and contaminated signals. Bar plots are given after computation.

Firstly, we tested the effect of different noises on the fast algorithm. Adding synthetic Gaussian noise to f_1 , f_2 and f_3 with zero

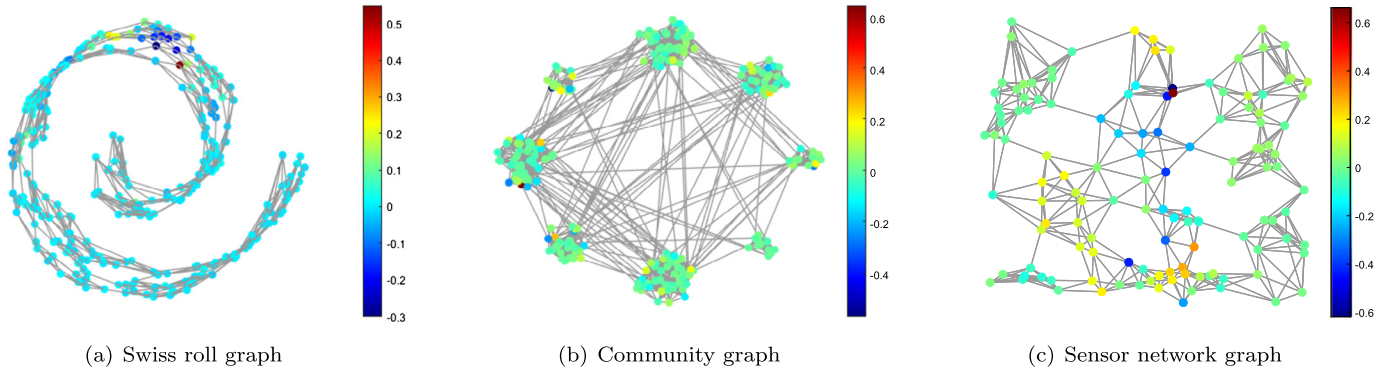
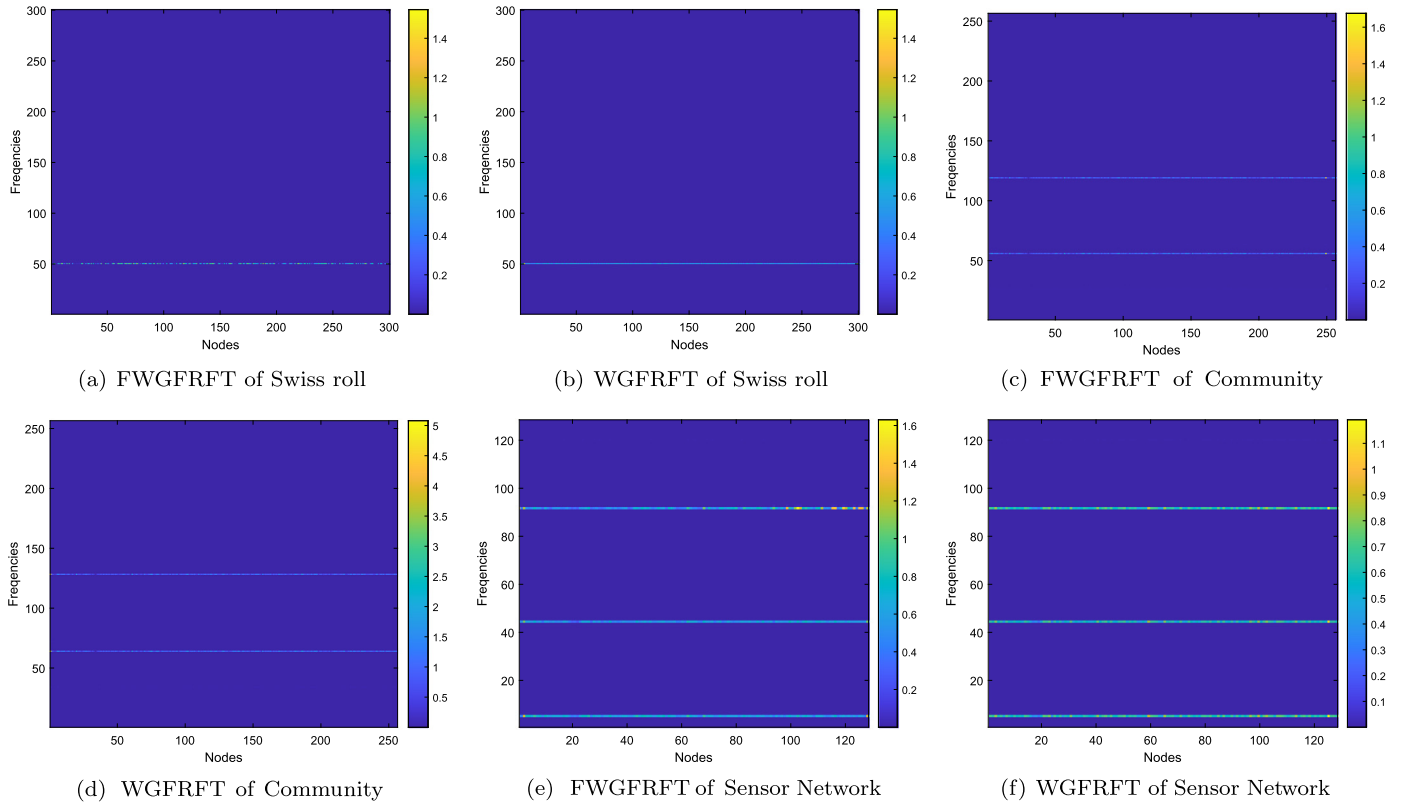


Fig. 3. Swiss roll, community, and sensor network graph.

Fig. 4. Vertex frequency representations of Swiss roll, Community, and Sensor graph by FWGRFT and WGRFT, with fractional order $\alpha = 0.9$.

mean and different variance σ . Fig. 5(a), Fig. 5(d), and Fig. 5(g) are the signals f_1 , f_2 and f_3 contaminated by Gaussian noise with $\sigma = 0.1$. In the first experiment, we add Gaussian noise with $\sigma = 0.2$ to signals on the path graph with 180 vertices and then apply FWGRFT, the fractional order is $\alpha = 0.9$. The vertex frequency representations are shown in Fig. 5(b), Fig. 5(e), and Fig. 5(h). Similarly, we add Gaussian noise with $\sigma = 0.5$ and the vertex frequency representations are shown in Fig. 5(c), Fig. 5(f) and Fig. 5(i). Compared with Fig. 2, we see that our algorithms are robust to Gaussian noise.

Secondly, we test FWGRFT with respect to different fractional orders. We select f_1 and f_3 as original signals and add Gaussian noises with different variances. The polluted signals are denoted by $\tilde{f}_1$ and $\tilde{f}_3$. Then we compute the root mean square error (RMSE) for the transformation coefficients, i.e.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i,l=1}^n \left((W_{\psi,\alpha} f)(i,l) - (W_{\psi,\alpha} \tilde{f})(i,l) \right)^2}. \quad (60)$$

From Fig. 6 we see the superiority of using fractional orders other than 1 (which is WGFT), since suitable WGRFT is robust to noise with large variances (see $\sigma = 0.2$ and $\sigma = 0.1$ in Fig. 6(a) and Fig. 6(b)). We remark that for small fractional orders ($\alpha = 0.1 \sim 0.3$), the RMSE is negligible since WGRFT is close to identity. Therefore, WGRFT is preferred for processing contaminated signals.

(ii) Time cost comparison

In order to show the time superiority of fast algorithm more clearly, we compared the time cost between the fast algorithm FWGRFT and the original algorithm WGRFT, setting fractional order $\alpha = 0.9$. We sample 20 ~ 320 points on the Swiss roll graph and use the signal $\text{Re}(\kappa_{N/2}^{\text{sr}})$, where N is the number of vertices, on the graph. We plot log-log figure where the x-axis represents the logarithm of the number of vertices and the y-axis represents the logarithm of calculation time (unit: second). The log-log figure is shown in Fig. 7. For WGRFT, red line fits the straight line with slope 4, which coincides with the order of theoretical compu-

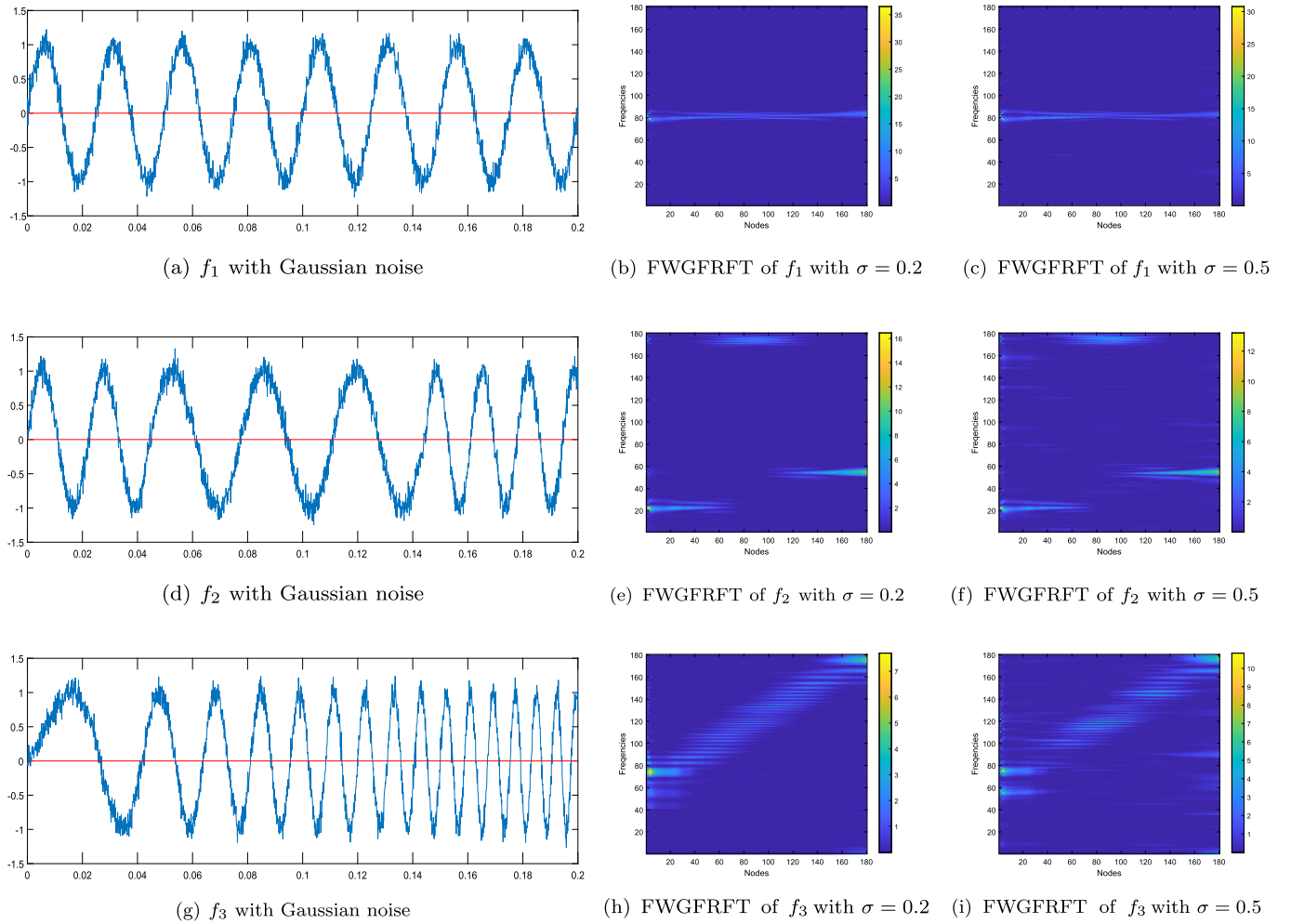


Fig. 5. The first three figures show signals polluted by Gaussian noise with variance $\sigma = 0.1$. Rest figures show vertex frequency representations with different variances after FWGFRFT.

tation complexity $O(N^4)$. Similarly, for FWGFRFT, the blue line fits the straight line with slope 3, which coincides with the order of theoretical computation complexity $O(N^3)$. From the results, it is obvious that fast algorithm is more efficient and requires less time than the original one.

6. Applications

We apply WGFRFT to cluster a graph into sets of vertices. In classical literature of statistical learning, the vertices are clustered according to the graph structure [37]. Our method utilizes the information contained in the signal. By using vertex frequency representation, vertices are clustered according to the affinity of vertex frequencies. We apply our method to two different graphs: one is the synthetic random sensor network, and the other is the real-world Minnesota road graph. The results show that our method can be well applied to the problem of graph clustering.

In the first experiment, we generate random signals on random sensor network with 512 vertices. In particular, three signals $\{f_i\}_{i=1,2,3}$ are generated with each component uniformly distributed between 0 and 1. Then we select filters as follows

$$\hat{h}_\alpha(\lambda) = \frac{1}{\sqrt{2\pi}\sigma} \exp^{-(\lambda-r_l)^2/2\sigma^2}, \quad (61)$$

where r_l denotes the fractional spectral of the sensor network. Fig. 8 shows four filters covering the whole graph fractional Lapla-

cian spectrum. We use the first three filters $\{\hat{h}_\alpha^l\}_{l=1,2,3}$ in the experiment.

Using shortest paths on the graph, we generate three clusters $\{C_i\}_{i=1,2,3}$ where each cluster is the metric ball of radius 3 around different center. The clusters are shown in Fig. 9(a). Next, we construct a signal f by

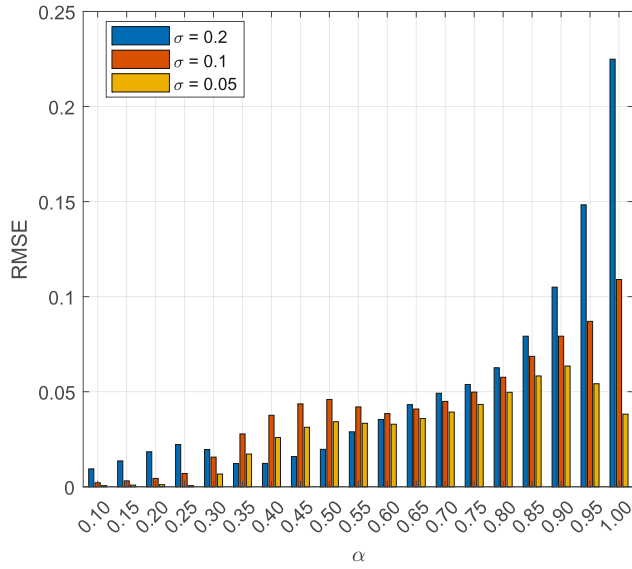
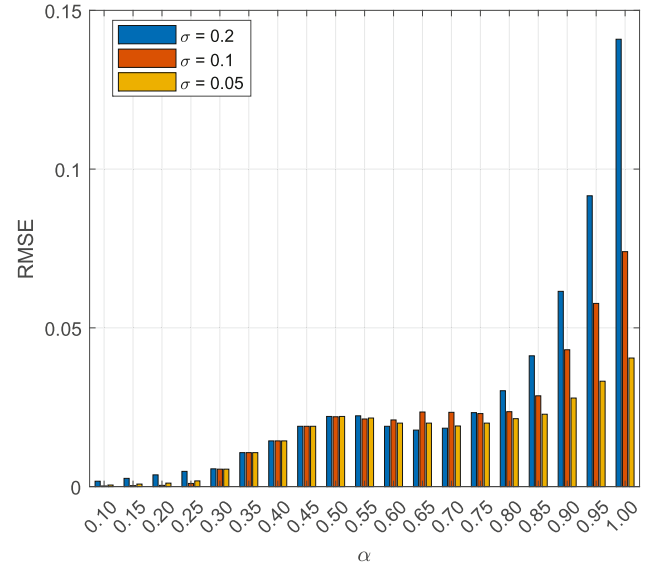
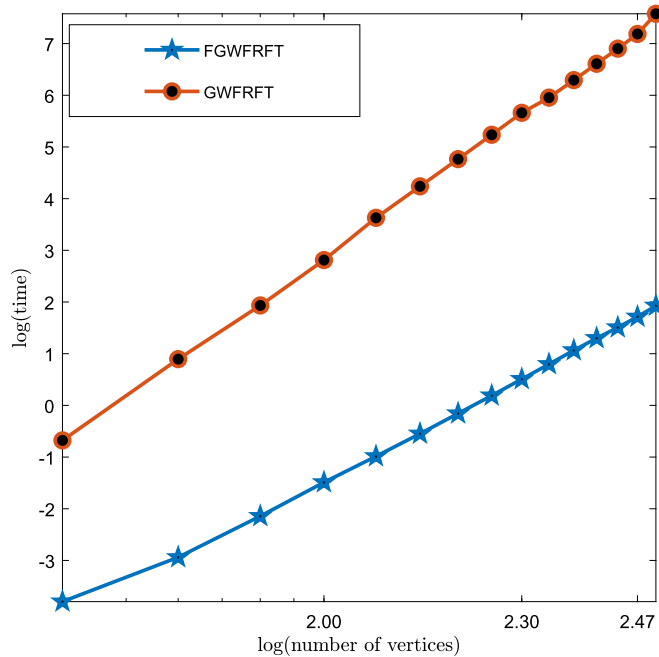
$$f_{net} = \sum_{i=1}^3 (f_i *_{\alpha} h_i)|_{C_i}. \quad (62)$$

That is, we filter each signal by $\{\hat{h}_\alpha^l\}$ and restrict to the cluster C_i . Outside the clusters the signal is set to be zero.

In the second experiment, we use the real-world Minnesota road graph. To show the independence of the selection of graph signals, we first embed graph nodes into 2-dimensional Euclidean space. Then we generate $\{C_i\}_{i=1,2,3}$ by setting C'_1 and C'_2 to be the nodes reside in the Euclidean ball of radius 1.5 and C'_3 to be the remaining points. The signals are still constructed using (62). However, in this case all nodes have nontrivial signals and there is no prior on the graph structure of the clusters.

In both cases, we apply WGFRFT to the signal f_{net} using Gaussian window. The vertex frequency representations are shown in Fig. 9(b) and Fig. 10(b). We see that there are three outstanding bands. Finally, we define the point cloud $\{y_i\}$ by

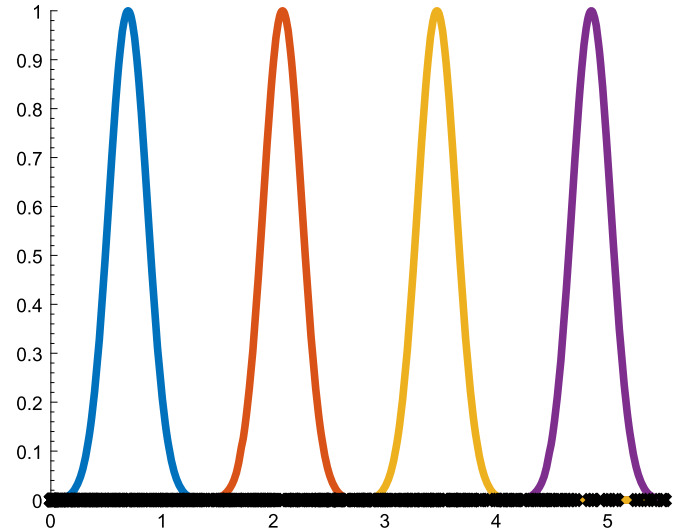
$$y_i(k) = s(W_{\psi,\alpha} f_{net})(i, k)^\beta, \quad (63)$$

(a) RMSE of $W_{\psi, \alpha} f_1$ and $W_{\psi, \alpha} \bar{f}_1$ (b) RMSE of $W_{\psi, \alpha} f_3$ and $W_{\psi, \alpha} \bar{f}_3$ **Fig. 6.** Bar plots of RMSE with respect to different fractional orders.**Fig. 7.** Time cost comparison of GWFRFT and FGWFRFT. The x-axis represents the logarithm of the number of vertices and the y-axis represents the logarithm of calculation time (unit: second).

where s and β are tuning parameters. In our experiment, we set $s = 10$ and $\beta = 0.4$ to sharpen the gaps of elements in $W_{\psi, \alpha} f_{\text{net}}$. The fractional order is $\alpha = 0.95$. We perform the standard k -means clustering on the point clouds $\{y_i\}$. The results are shown in Fig. 9(c) and 10(c). We can see in both cases our method successfully recognizes the true clusters.

7. Conclusions

A framework for constructing and calculating a novel transform on weighted graphs is proposed. Using graph fractional Laplacian eigenvectors, three graph fractional operators and related properties are presented. Then a new transform called windowed graph

**Fig. 8.** Four filters $\{\hat{h}_\alpha^l\}_{l=1,2,3,4}$.

fractional Fourier transform (WGFRFT) is designed. Using windowed graph fractional Fourier coefficients, a signal reconstruction formula as well as the inverse transform is shown. A property of the window function guarantees the reasonableness of this inverse transform. To improve the computation time, a fast algorithm for WGFRFT is shown. The correctness and efficiency are verified on synthetic data sets. Finally, an application of WGFRFT to graph clustering is also given to highlight its potential utility.

CRediT authorship contribution statement

Fang-Jia Yan: Conceptualization, Methodology, Software, Writing – original draft. **Bing-Zhao Li:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

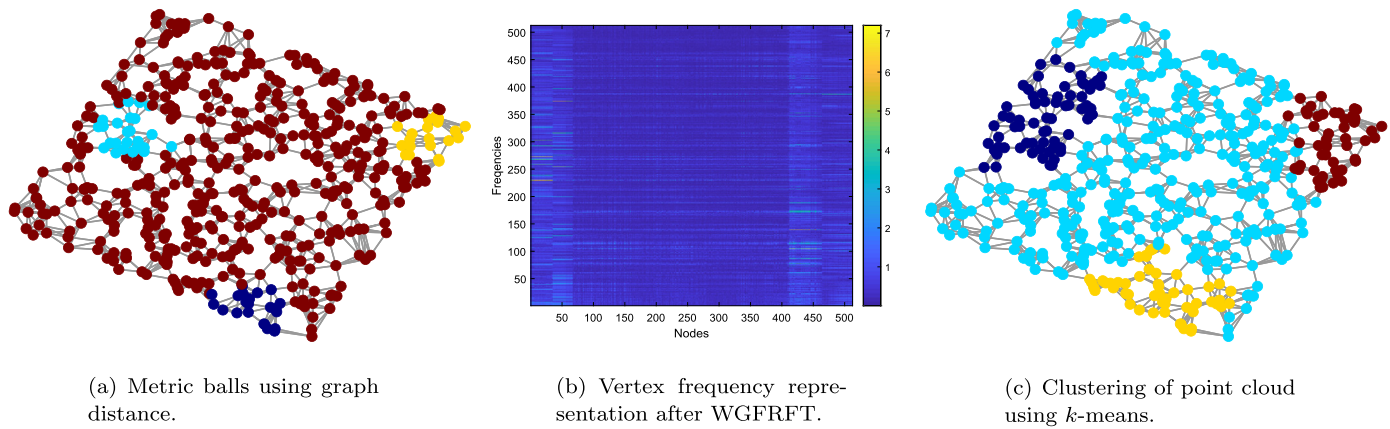


Fig. 9. Graph signal clustering process for random sensor networks.

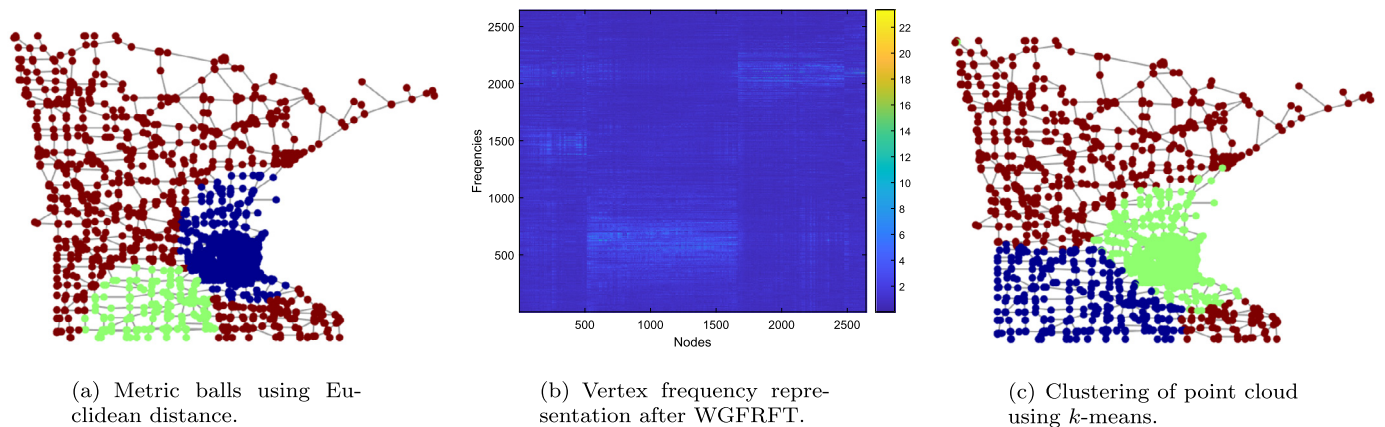


Fig. 10. Graph signal clustering process for Minnesota road graph.

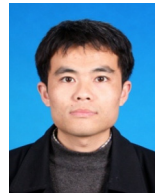
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